

“Looking Back: Sentiment Analysis of Elon Musk’s Wikipedia Page Over 10+ Years”

CPT 127 assignment - “Intro to Sentiment Analysis using Python”

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Note: accompanying code can be found at:

https://github.com/sschlotterbeck/cpt_127_assignments/blob/intro_to_sentiment_analysis_assignment/sentiment_analysis_proj.py

https://github.com/sschlotterbeck/cpt_127_assignments/blob/intro_to_sentiment_analysis_assignment/Interactive%20-%20sent_analy_proj_plot_interactive.py.ipynb

https://github.com/sschlotterbeck/cpt_127_assignments/blob/intro_to_sentiment_analysis_assignment/sentiment_analysis_urls.csv

Introduction

For this project, which introduced us to sentiment analysis in python, I chose to analyze the Wikipedia entry for Elon Musk. I scraped archived versions of Elon Musk’s Wikipedia entry from the saved snapshots on archive.org’s “Wayback Machine” twice a year (January and June) starting in 2015 and continuing til June of this year (2025), for a total of 22 entries to analyze.

I originally chose this data because I heard that Wikipedia had generous and clear policies on scraping their data, and was looking for an angle that would make sense to do sentiment analysis on with Wikipedia. That led me to the idea of comparing a single wiki as the years went by, which led me to the Wayback Machine (and not actually requesting web content from Wikipedia itself after all).

Elon Musk popped into my head as a potentially interesting wiki to investigate, partially because he is such a prominent and controversial figure in general, partially because his actions and words have becoming especially visible and controversial in 2025, and partially because both his public comments on wikipedia have varied so widely over the years, and he has complained about alleged bias on the part of the wikipedia editors in editing his entry. I figured that his wikipedia entry had as much chance any that I could think of, to have been subject to intense scrutiny and debate amongst the Wikipedia editors and both pro- and anti-Musk contingents of people, and potentially have interesting things to show from sentiment analysis.

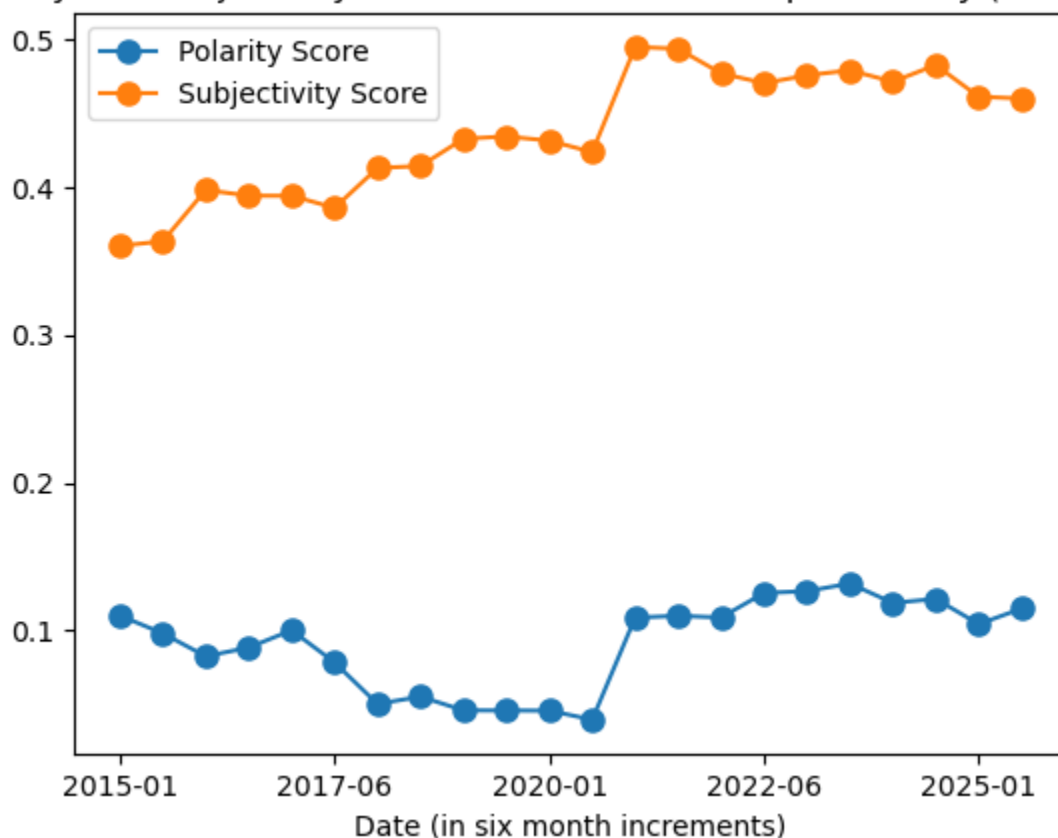
Process

After requesting each page with the requests library, I extracted and parsed the body of the page with BeautifulSoup, and then passed that text to TextBlob for sentiment analysis, specifically requesting the scores for Polarity and Subjectivity, which I added to separate lists called “polarities” and “subjectivities”.

I set up this process as a loop to be repeated for each URL address in the csv file. I was having some trouble with the requests timing out when I ran the code all at once, (even when I added a time.sleep() line at the end of the loop), so I eventually came up with a line that prompts the user to hit “Enter” each time one iteration of the loop completes to run the next iteration. I also added a counter to the loop, so the user can keep track of how many more pages there are to process.

Once I got the Polarity and Subjectivity scores for each archived version of the page, I placed them into a DataFrame and plotted the scores over time onto a graph, which you can see on the next page.

Polarity and Subjectivity Score of Elon Musk's Wikipedia entry (2015-2025)



Reflections and Conclusions

I really enjoyed the discovery process of getting to see how the polarity and subjectivity scores¹ fluctuated over the years. I went into this having no idea what it would look like, or even if it would change much at all. Looking at the graph now makes me very curious to investigate what might have happened in the Wikipedia page and in real life to cause such a shift in both the polarity and subjectivities scores in January of 2021—increasing the relative positivity as well as the subjectivity of the entry. I think the main thing that surprised me was just how just these 44 points on a graph started to suggest whole worlds of meaning, reasons, stories hidden behind them—I was genuinely surprised to just look at the graph and begin to imagine what the reasons for the changes might be. This in turn made me curious about analyzing other Wikipedia entries in a similar manner, and whether such archival investigation is already done, and how it could be made use of in the future.

¹ I used TextBlob (<https://textblob.readthedocs.io/en/dev/>) to process the entries for polarity and subjectivity scores. The polarity score scale used in TextBlob ranges from -1.0 to 1.0, with -1.0 representing very negative sentiment and 1.0 representing very positive sentiment. The subjectivity score ranges from 0.0 (highly objective) to 1.0 (highly subjective).